TRUONG VAN TINH

AI Application Engineer

Python · FastAPI · RAG · LLM Integration · Evaluation

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PROFESSIONAL SUMMARY

Software engineer applying AI in production retail work at Con Cung (vector semantic search, n8n market-intelligence pipelines, a Claude + MCP debugging loop) and building RAG question-answering with citations and LLM ticket-triage automation in Python and FastAPI, with validated outputs, fallbacks and automated evaluation. 3+ years of software development, including .NET, SQL Server and Keycloak.

WORK EXPERIENCE

Full Stack Developer at Con Cung

Nov 2023 – Present

Vietnam's mother-and-baby retail chain · .NET · AngularJS · SQL Server · Dapper · Keycloak · n8n · Claude + MCP

- Built vector semantic search into the company's commerce portal, retrieving products and content by meaning rather than keywords.
- Automated competitor-data scraping with n8n pipelines that feed market-intelligence reporting.
- Engineered an autonomous debug-and-test loop with Claude, Agent Skills and MCP; used Custom GPTs and prompt engineering to speed up delivery.
- Designed retail data structures for shelves, facings and display groups behind store-layout optimization.
- Delivered AngularJS and .NET voucher workflows, 7 subscription benefit calculations and a tool for 4 generated code-template types.
- Implemented Keycloak JWT / OAuth 2.0 / OIDC authentication and role-based authorization; supported Docker and IIS deployment.
- Fixed SQL Server / Dapper retrieval defects and verified ERP API request and response mappings.

Frontend Intern at FPT

Sep 2023 – Nov 2023

- Developed Angular / JavaScript components, integrated e-commerce REST APIs and corrected mobile UI defects.

PROJECTS

Retail RAG Assistant — PERSONAL PROJECT

Python, FastAPI, LangChain, Qdrant, TF-IDF, Docker

github.com/tvt987/retail-rag-assistant

- Built document parsing and preprocessing for Markdown, HTML and PDF (Unicode normalization, boilerplate removal) into cited chunks.
- Stored TF-IDF sparse or SentenceTransformers dense vectors in a Qdrant vector database; added a LangChain LCEL pipeline.
- Developed FastAPI endpoints with low-score abstention and extractive fallback when LLM calls fail or answers lack citations.
- Evaluated 16 questions on hit rate, MRR, fact recall and groundedness that rejects wrong numbers and uncited claims; 29 tests in CI.

Support Ticket AI Triage — PERSONAL PROJECT

Python, FastAPI, LLM Integration, SQLite, Docker

github.com/tvt987/support-ticket-ai-triage

- Automated LLM-assisted triage for 5 ticket categories with priority and sentiment validation, urgent-rule overrides and a rules fallback.
- Protected human review with a token, masked contact data and kept raw ticket text out of storage; 5 synthetic fixtures passed.
- Tested 10 FastAPI HTTP scenarios with mocked LLM responses, 120 schema combinations and 720 urgent-rule cases; verified SQLite persistence for 1,000 review decisions.

Retail Reliability Lab — AI-ASSISTED LAB

FastAPI, Redis, SQLite, Docker Compose, Azure

- Validated Redis cache-aside with a 60-second TTL, invalidation and outage fallback; 22 API and cache tests against real Redis.
- Cut median SQLite lookup 3.63 ms → 0.005 ms (200 queries, 100,000 synthetic rows); authored Docker Compose and Azure templates.

SKILLS AND TECHNOLOGIES

AI applications	RAG, Document Retrieval, Data Pipelines, NLP (text retrieval), TF-IDF, Semantic Embeddings, LLM Integration, Prompt Engineering, Output Validation, Evaluation, Abstention and Fallbacks, Groundedness Evaluation, Vector Search, Vector Databases, Human-in-the-loop Workflows
Python & APIs	Python, FastAPI, REST APIs, SQLite, Input Validation, Document Parsing, Data Preprocessing
Backend & data	C#, .NET, ASP.NET Core, Dapper, SQL Server, PostgreSQL, Redis, Database Design, Query Optimization
Security	Keycloak, JWT, OAuth 2.0, OpenID Connect, RBAC, Contact-data Masking
Testing & DevOps	Unit Testing, Integration Testing, Docker, CI/CD, GitHub Actions, Git, Logging, Monitoring/Observability, Azure, AWS, IIS
Tools & frontend	LangChain, Qdrant, AI Agents, OpenAI (ChatGPT, Custom GPTs), Claude, Cursor, GitHub Copilot, MCP, n8n, Angular, AngularJS, TypeScript, JavaScript, RxJS, React, HTML5, CSS3, Tailwind CSS
Practices	Agile/Scrum, Code Review, Technical Documentation, Problem Solving, Cross-team Collaboration, Communication

EDUCATION — **University of Information Technology (UIT), VNU-HCM** · Information Technology · Graduated 2026

LANGUAGES — **English** Good reading and writing (technical documents); basic spoken communication